J Clin Monit Comput (2017) 31:59–65
DOI 10.1007/s10877-015-9822-4

ORIGINAL RESEARCH

Acoustic method respiratory rate monitoring is useful in patients under intravenous anesthesia

Kentaro Ouchi[1] · Shigeki Fujiwara[2] · Kazuna Sugiyama[3]

Received: 18 September 2015 / Accepted: 29 December 2015 / Published online: 12 January 2016
© Springer Science+Business Media Dordrecht 2016

Abstract Respiratory depression can occur during intravenous general anesthesia without tracheal intubation. A new acoustic method for respiratory rate monitoring, RRa® (Masimo Corp., Tokyo, Japan), has been reported to show good reliability in post-anesthesia care and emergency units. The purpose of this study was to investigate the reliability of the acoustic method for measurement of respiratory rate during intravenous general anesthesia, as compared with capnography. Patients with dental anxiety undergoing dental treatment under intravenous anesthesia without tracheal intubation were enrolled in this study. Respiratory rate was recorded every 30 s using the acoustic method and capnography, and detectability of respiratory rate was investigated for both methods. This study used a cohort study design. In 1953 recorded respiratory rate data points, the number of detected points by the acoustic method (1884, 96.5 %) was significantly higher than that by capnography (1682, 86.1 %) ($P < 0.0001$). In the intraoperative period, there was a significant difference in the LOA (95 % limits of agreement of correlation between difference and average of the two methods)/ULLOA (under the lower limit of agreement) in terms of use or non-use of a dental air turbine ($P < 0.0001$). In comparison between capnography, the acoustic method is useful for continuous monitoring of respiratory rate in spontaneously breathing subjects undergoing dental procedures under intravenous general anesthesia. However, the acoustic method might not accurately detect in cases in with dental air turbine.

Keywords Physiological monitoring · Respiratory complications · Intravenous general anesthesia · Dental operation

Clinical Trial Registration: UMIN No. UMIN000016417.

✉ Kentaro Ouchi
ken2006anes@yahoo.co.jp

[1] Department of Dental Anesthesiology, Field of Maxillofacial Diagnostic and Surgical Sciences, Faculty of Dental Science, Kyushu University Graduate School, 3-1-1 Maidashi, Higashi-ku, Fukuoka 812-8582, Japan

[2] Department of Anesthesiology, Graduate School of Dentistry, Kanagawa Dental University, 82 Inaoka-cho, Yokosuka, Japan

[3] Department of Dental Anesthesiology, Field of Oral and Maxillofacial Rehabilitation, Kagoshima University Graduate School of Medical and Dental Sciences, 8-31 Sakuragaoka, Kagoshima, Japan

1 Introduction

In dental practice, intravenous anesthesia without tracheal intubation is often used for patients who are difficult to treat due to dental anxiety [1–3]. However, since respiratory depression due to airway obstruction can occur in these patients, perioperative respiratory monitoring is very important. Several respiratory rate monitoring techniques are available. One such technique, capnography, is often used because of its ability to detect cessation of breathing before there is a decrease in pulse oxygen saturation. However, since capnography requires the use of a sampling line connected to an oral or nasal mask, its use during dental treatment is limited. A new acoustic method for respiratory rate monitoring (RRa®, Masimo Corp., Tokyo, Japan) has shown good reliability when used in extubated patients in post-anesthesia care units [4, 5], intensive care units [6], and emergency units [7]. In this method, the device probe is connected to the neck, and hence, it does not interfere with dental treatment. Therefore, the acoustic method may be

useful during intravenous anesthesia for dental treatment as compared with other methods of respiratory rate monitoring.

The purpose of this study was to determine the reliability of the acoustic method as compared with capnography in patients undergoing dental procedures under intravenous general anesthesia.

2 Methods

This observational study was conducted with the approval of the ethics committee of Kagoshima University Medical Dental Hospital in September 2014 (No. 466), in accordance with the STROBE statement for reporting observational studies and the principles of the Ethical Guidelines for Clinical Studies of the Ministry of Health, Labor and Welfare in Japan. It was registered with the University Hospital Medical Information Network Clinical Trials Registry (number UMIN000016417), which was established for national medical schools in Japan.

Patients with dental anxiety undergoing dental treatment under intravenous anesthesia without tracheal intubation, aged 18 years or older and ASA PS 1 (Normal healthy patient), were enrolled in this study, after obtaining their written informed consent for dental treatment and participation in this study under intravenous general anesthesia.

2.1 Intravenous general anesthesia protocol

The patients did not receive premedication. Patients were monitored during anesthesia [blood pressure, pulse oximetry, and respiration (described below)]. Intravenous general anesthesia was induced with a continuous infusion of propofol using a target-control infusion (TCI) pump (TERUFUSION TE-371, Terumo Co.) with a built-in TCI system (Diprifusor, AstraZeneca Co.) and according to the parameters reported by Marsh [8]. The dose of propofol was titrated to achieve a Mackenzie Grant score of 5 and to achieve an adequate level of anesthesia: asleep, but not responding to stimulation. Tracheal intubation was not performed and spontaneous breathing was maintained. The level of anesthesia was maintained at a Mackenzie and Grant score of 5 by adjusting the target propofol level using the propofol TCI (Fig. 1). If respiratory depression was observed (snore and/or paradoxical breathing), the target blood concentration of propofol was decreased by 0.2 μg/ml. If the anesthesia level was deemed inadequate, the target blood level of propofol was increased by 0.2 μg/ml. The dental procedure was started after the anesthesia level became stable without respiratory depression. A local anesthetic (2 % lidocaine with 1:80,000 adrenaline) was used as appropriate. Administration of propofol was discontinued at the end of the dental procedure. Patients were monitored until recovery from anesthesia, when they were fully awake and had stable respiration.

2.2 Monitoring of respiration

Monitoring of respiratory rate was done using two methods. The acoustic method of respiratory rate measurement involved use of an adhesive sensor, including an acoustic transducer (RAS-125, Masimo Corp., Tokyo, Japan), placed on the patient's throat, on the side of the larynx and above the thyroid cartilage, and connected to a specific device (Radical-7 Pulse Oximeter, Masimo Corp., Tokyo, Japan). The sensor allows real time analysis of the vibrations emanating from the patient's larynx and enables isolation of the respiratory sounds via analysis of the trace obtained through various filters. The acoustic signal is then converted to a numerical value, allowing a continuous display of the respiratory rate value. Measurement of respiratory rate by capnography involved the use of a nasal cannula (Psychorich IS cannula, Sekimura Corp, Tokyo, Japan) connected to a carbon dioxide sampling unit and a standard automated monitoring device (BP-608, Omron Colin, Tokyo, Japan). Both devices continuously recorded the subjects' respiratory rates throughout the perioperative period.

2.3 Measurement of parameters

The respiratory rates indicated by the two devices were recorded every 30 s, from start to end of anesthesia. First, to correctly set the time of the clock of the two devices. Second, the respiratory rates recorded automatically by the two devices. Third, the respiratory rates extracted every 30 s (0 and 30 s in a minute) from automatically recording data by the two devices.

2.4 Statistical analyses

The number of detected data points by the acoustic method compared with capnography over the observation period, from the start to end of anesthesia, were compared for the following periods: anesthetic dose commencement to start of dental procedures (anesthesia induction period), start to end of dental procedures (intraoperative period), and end of dental procedures to end of anesthesia (anesthesia emergence period).

Respiratory rate that were recorded by both the acoustic method and capnography were compared with each other. Comparison of respiratory rate values obtained by the two methods was assessed by the Bland and Altman method, with the calculation of bias and of 95 % limits of agreement (LOA). Next, the correlation between differences in respiratory rate (acoustic method − capnography) and average respiratory rate of the two methods [(acoustic method + capnography)/

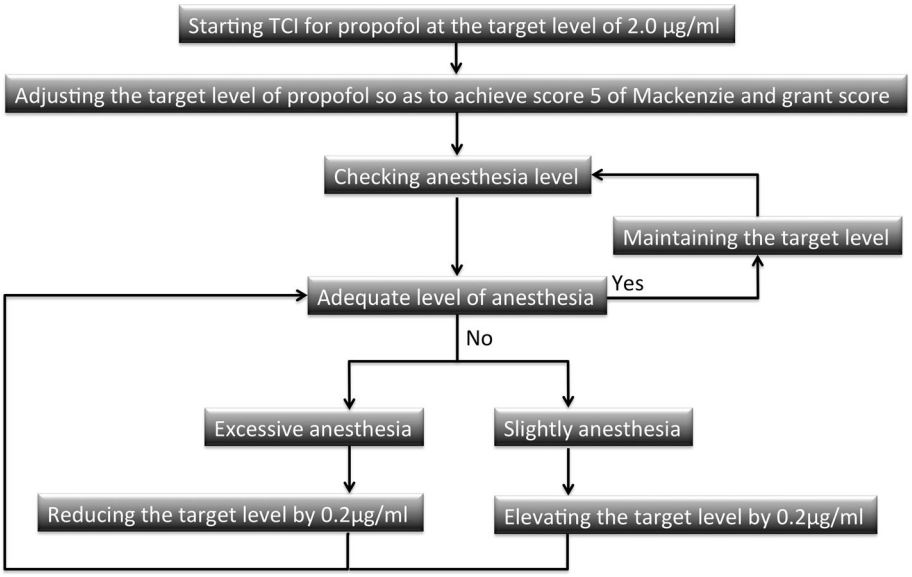

Fig. 1 Intravenous general anesthesia protocol. *TCI* target-controlled infusion

2] was evaluated. Bias between the two methods (acoustic method − capnography) was compared between the three observation periods. To evaluate the effect of use of a dental air turbine on measurements by the acoustic method, the ULLOA (under the lower limit of agreement)/LOA in Bland–Altman analysis, i.e. ULLOA at which the respiratory rate measured by the acoustic method was lower than that by capnography, was compared with use/non-use of the dental air turbine for dental procedures.

For comparison of detectable respiratory rate data points and LOA/ULLOA, the Chi-square test was employed. For comparison of continuous variables between the three groups, two-way ANOVA and the Tukey test were employed. JMP software (version 10, SAS Institute Inc., Japan) was used for statistical analysis, and $P < 0.05$ was regarded as statistically significant. The results are presented as the absolute value (percentage) for categorical variables and as the mean ± standard deviation (SD) for continuous variables.

3 Results

Eleven patients were included in this study. The patients included 4 men and 7 women, 38 ± 13.8 (mean ± SD) years of age, 162.2 ± 7.3 cm in height, 64.7 ± 13.4 kg in weighing, and 115.7 ± 26.3 min in anesthesia time. The dental procedures performed included endodontics without dental air turbine (n = 6), tooth extraction with dental air turbine (n = 3), impression taking without dental air turbine (n = 1), and crown preparation with dental air turbine (n = 1). A total of 1953 data points of respiratory rate were obtained from the start to end of anesthesia.

Of these 1953 recorded points, the detectability of respiratory rate by the acoustic method over the entire observation period was significantly higher than that by capnography ($P < 0.0001$) (Table 1). In the anesthesia induction period, the detectability of respiratory rate by the two methods was not significantly different ($P = 0.433$); in the intraoperative period, the acoustic method was significantly higher than capnography ($P < 0.0001$); in the anesthesia emergence period, the acoustic method was significantly higher than capnography ($P = 0.030$) (Table 2). The detectability of respiratory rate during the three periods was compared for each method. With both the acoustic method and capnography, the detectability of respiratory rate was different between the three periods ($P < 0.0001$ and $P = 0.0002$, respectively).

Of the 1953 data points obtained during the entire observation period in all 11 patients, 1636 recorded points were detected by both the acoustic method and capnography. Bland–Altman plots comparing the difference in respiratory rate values by the acoustic method and capnography (acoustic method − capnography) during anesthesia induction, the intraoperative period and anesthesia emergence period are

Table 1 Respiratory rate data points [n (%)] over the entire observation period

	Capnography		
	Detected	Not detected	
Acoustic method			
Detected	1636 (83.7)	248 (12.7)	1884 (96.5)
Not detected	46 (2.4)	23 (1.2)	69 (3.5)
	1682 (86.1)	271 (13.9)	1953 (100)

"Not detected" indicated that respiratory monitoring hardware detected apnea, each respiratory monitoring method

$P < 0.0001$

Table 2 Respiratory rate data points [n (%)] detected by the two methods in each period

	Anesthesia induction period		Intraoperative period		Anesthesia emergence period	
	Detected	Not detected	Detected	Not detected	Detected	Not detected
Acoustic method	210 (91.3)	20 (8.7)	1407 (97.2)	41 (2.8)	267 (97.1)	8 (2.9)
Capnography	205 (89.1)	25 (10.9)	1221 (84.3)	227 (15.7)	256 (93.1)	19 (6.9)
P value	$P = 0.433$ (NS)		$P < 0.0001$		$P = 0.030$	

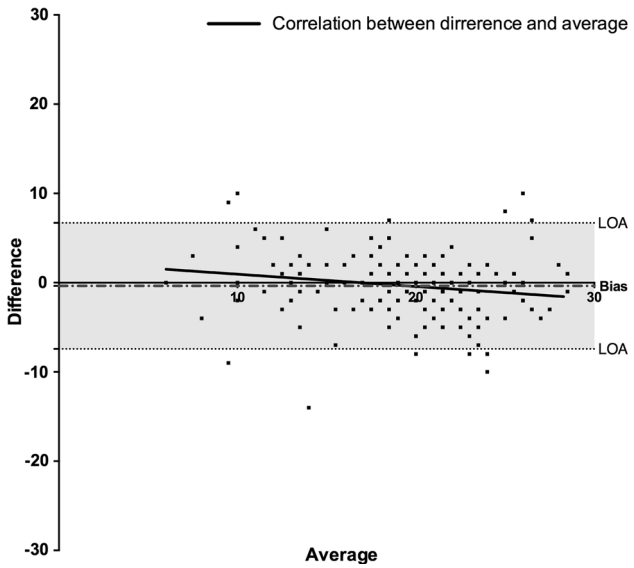

Fig. 2 Bland–Altman analysis (acoustic method − capnography) of respiratory rate measured by the two methods in the anesthesia induction period. Comparison of respiratory rate values obtained by the two methods was assessed by the Bland and Altman method, with the calculation of bias and of 95 % limits of agreement (LOA). Bias = −0.3636, LOA −7.424 to 6.696, $R^2 = 0.03253$

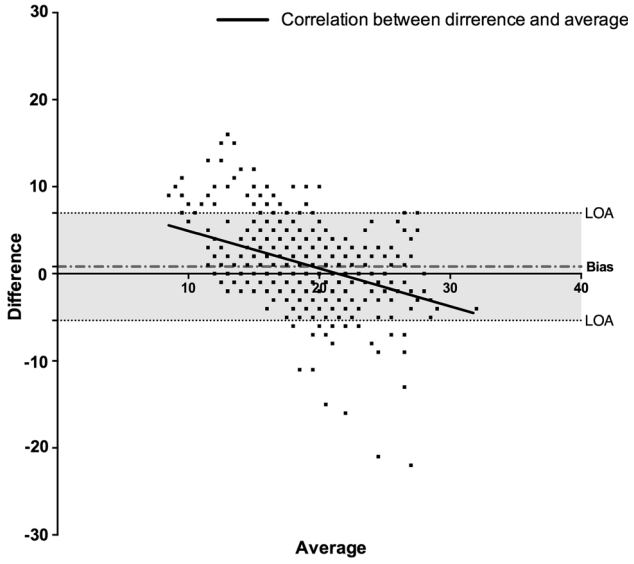

Fig. 3 Bland–Altman analysis (acoustic method − capnography) of respiratory rate indicated by the two methods in the intraoperative period. Bias = 0.8067, LOA −5.353 to 6.966, $R^2 = 0.1799$

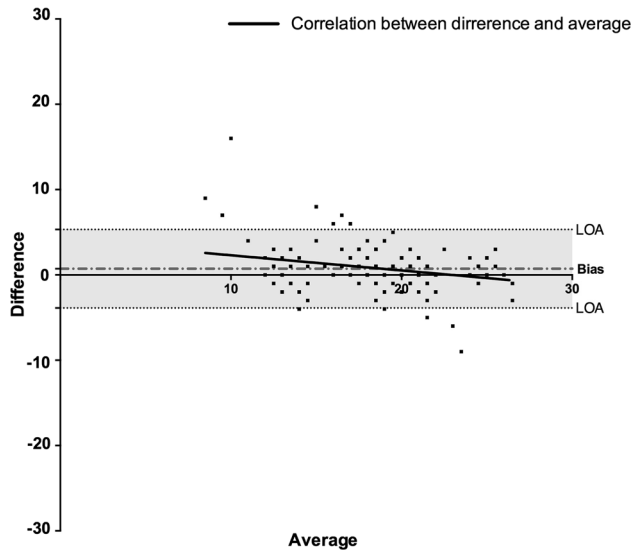

Fig. 4 Bland–Altman analysis (acoustic method − capnography) of respiratory rate indicated by the two methods in the anesthesia emergence period. Bias = 0.7229, LOA −3.877 to 5.323, $R^2 = 0.07228$

Table 3 Comparison of bias of respiratory rate values (beat/min) between the acoustic method and capnography (acoustic method − capnography), each period

	Bias (limits of agreement)
Anesthesia induction period	−0.36 ± 0.23 (−7.424 to 6.696)
Intraoperative period	0.81 ± 0.09 (−5.353 to 6.966)
Anesthesia emergence period	0.72 ± 0.20 (−3.877 to 5.323)

Anesthesia induction period versus intraoperative period, $P < 0.0001$

Anesthesia induction period versus anesthesia emergence period, $P = 0.0009$

presented in Figs. 2, 3, and 4, respectively. Comparison of the bias between the two methods (acoustic method − capnography) in the three observation periods indicated that the bias was different between the three periods ($P < 0.0001$) (Table 3).

Next, ULLOA/LOA was compared with and without the use of a dental air turbine for dental procedures in each period (Tables 4). There was no significant difference in the ULLOA/LOA in terms of use or non-use of a dental air turbine in the anesthesia induction period and anesthesia

Table 4 Use of a dental air turbine for surgery and the number of LOA/ULLOA data points [n (%)] in each period

	Anesthesia induction period		Intraoperative period		Anesthesia emergence period	
	Use	Non-use	Use	Non-use	Use	Non-use
LOA	90 (52)	84 (48)	262 (23)	853 (77)	85 (36)	148 (64)
ULLOA	5 (71)	2 (29)	17 (74)	6 (26)	3 (50)	3 (50)
P value	$P = 0.4481$ (NS)		$P < 0.0001$		$P = 0.6721$ (NS)	

emergence period. In the intraoperative period, there was a significant difference in the ULLOA/LOA in terms of use or non-use of a dental air turbine ($P < 0.0001$).

4 Discussion

There was the report using an acoustic method for respiratory rate monitoring in such as post-anesthesia care units, intensive care units, and emergency units. This study is the first study evaluating respiratory rate monitoring by an acoustic method during intravenous anesthesia for dental procedures. Moreover, this study is the first to assess the detectability of respiratory rate and its reliability during different phases of the anesthesia procedure.

Pulse oximetry is usually used for monitoring of respiratory function during intravenous anesthesia. However, it has certain limitations. Under anesthesia with oxygen supplementation, the decrease in pulse oxygen saturation may be delayed despite severe respiratory insufficiency [9, 10]. Respiratory rate monitoring detects early respiratory distress [4, 11, 12]. Thus, it is recommended that monitoring during intravenous anesthesia should include both respiratory rate monitoring as well as pulse oximetry. Several respiratory rate monitoring techniques are available. Capnography using a nasal or facial interface is a useful technique [13–15]. Recently, the acoustic method for respiratory rate monitoring has been used in post-anesthesia care units, intensive care units, and in the operating room under laryngeal mask airway general anesthesia [4, 5, 16–18]. Thus, we compared the acoustic method with capnography during non-intubated intravenous general anesthesia for dental procedures. In previous studies, assessment of respiratory rate by the acoustic method was compared with capnography coupled with a face mask [7]. During intravenous anesthesia for dental procedures, however, a facemask is not used. Therefore, capnography coupled with a nasal cannula was assessed in this study. Although respiratory monitoring by capnography coupled with a nasal cannula has the drawback of being inaccurate in the event of mouth breathing, this technique is commonly used in the clinical practice of intravenous anesthesia for dental procedures and oral surgery.

The detectability of respiratory rate by the acoustic method compared with by capnography was assessed. In result, 248 points (12.7 %) could be not detected by capnography while acoustic method could detect. On other hand, 46 points (2.4 %) could be not detected by acoustic method while capnography could detect. This indicates that, the patient's breathing changed to mouth breathing, as results capnography could not detect at 12.7 %. And, in 2.4 % in capnography could detect; it is indicated that, though there is not effective breathing, capnography detected or, alternatively though there is effective breathing, acoustic method could be not detected. Nevertheless, our results indicated that the detectability power of the acoustic method was better than that of capnography over the entire observation period. In particular, the respiratory rate detectability of capnography during the intraoperative and anesthesia emergence periods (84.3 and 93.1 % detected, respectively), after titration of the anesthetic and stabilization of anesthesia depth, was worse than that of the acoustic method (97.2 and 97.1 % detected, respectively). This indicates that the patient's breathing often changed to mouth breathing under anesthesia. This suggests that the superiority of the acoustic method over capnography via a nasal cannula for respiratory rate monitoring during intravenous anesthesia is due to its ability to detect respiration even if the patient starts mouth breathing, as often occurs under intravenous anesthesia. Hence, the acoustic method is a useful monitoring technique during dental surgery under intravenous general anesthesia. During the anesthesia induction period, respiratory rate detectability was not different between the two methods in this study. During this period, the anesthesiologist performs titration of the anesthesia level to achieve stable anesthesia without respiratory depression. Yet, respiratory depression sometimes occurs during this period. This could indicate that when respiratory depression is caused by anesthesia induction, both methods are not able to assess the respiratory rate.

We also compared the number of detected data points with each method between the 3 periods: anesthesia induction period, intraoperative period, and anesthesia emergence period. With use of the acoustic method in the anesthesia induction period, namely, during commencement of administration of the anesthetic to stabilization of the anesthesia level, the respiratory rate detectability was lower than during the other two periods. Anesthesia is assessed to be stable when there is no response to stimulation and there is no respiratory depression/arrest. During the process of anesthetic dose titration to achieve stable anesthesia, respiratory arrest

sometimes occurs. At this time, if only capnography is used for respiratory monitoring, it may be difficult for the anesthesiologist to determine whether the reason for the absent respiratory rate measurement is respiratory arrest or a change to mouth breathing. Therefore, the acoustic method is a useful respiratory rate monitoring method during the period from commencement of anesthetic administration to stabilization of the level of anesthesia. Respiratory rate detectability by the acoustic method was higher than that by capnography over the entire observation period, indicating that nasal capnography often did not accurately detect respiration because of a change to mouth breathing. Measurement errors with capnography can also occur due to specific events, such as speaking and moving [4]. In this study, speaking and moving did not affect the results since the level of anesthesia was adequate to prevent such events.

Comparison of bias in the respiratory rate values between the two methods (acoustic method − capnography) over the three observation periods indicated a greater bias during the intraoperative period as compared to the anesthesia induction period and during the intraoperative period as compared to the anesthesia emergence period. It considered that the greater bias during the two periods with large biases could be because more respirations were detected by the acoustic method than by capnography because of a change to mouth breathing at these times, because the respiratory rate detectability power by the acoustic method was significantly higher than that by capnography in the two periods with large biases. This suggests that the respiratory rate values indicated by capnography are inaccurate when change to mouth breathing. Therefore, the acoustic method is a useful respiratory rate monitoring method when the patient's breathing changes to mouth breathing.

In this study, the correlation between differences in respiratory rate value (acoustic method − capnography) and average respiratory rate value [(acoustic method + capnography)/2] was assessed for each period. R^2 of the correlation in the intraoperative period ($R^2 = 0.1799$) was larger than in the other two periods (0.03253 and 0.07228 in the induction and emergence periods, respectively). This difference in R^2 in the intraoperative period in contrast to the other two periods suggests that in cases in which the average respiratory rate value by both methods is small, a higher respiratory rate value is indicated by the acoustic method, and in cases in which the average rate is large, a lower respiratory rate value is indicated by the acoustic method than by capnography. In the former situation, respiratory rate value by capnography was probably low because of mouth breathing, although the acoustic method accurately detected the rate. In the latter situation, respiratory rate value was accurately detected by capnography but not by the acoustic method. Respiratory rate value by the acoustic method is calculated using the sound of breathing. Therefore, we compared the ability of the two methods to accurately determine the respiratory rate value with and without the use of a dental air turbine for dental procedures. Cases that were ULLOA (under the lower limit of agreement) in Bland–Altman analysis included those in which the respiratory rate value was accurately detected by capnography but not by the acoustic method. We assessed ULLOA/LOA with the use/non-use of a dental air turbine in each period. During the anesthesia induction and emergence periods, namely when there was no dental procedure being performed, there was no difference in ULLOA/LOA with and without the use of the dental air turbine. This indicates the appropriateness of the assessment method, because the dental air turbine was not used during these two periods. In contrast, in the intraoperative period, ULLOA with use of the air turbine being significantly higher than ULLOA with non-use of the dental air turbine, and LOA with use of the dental air turbine being significantly lower than LOA with non-use of a dental air turbine. Therefore, the acoustic method does not accurately detect respiratory rate value in cases in which a dental air turbine is used during the procedure.

Our study has several limitations. As mentioned above, the acoustic method has a lower reliability in cases in which the dental air turbine is used. However, in such cases, the operating dentist does not use the air turbine throughout the dental procedure. If we had examined the effect of use of the turbine by dividing such cases into turbine use and non-use periods, we may have been able to precisely examine the influence of the air turbine. An other limitation of our study is that this study measured respiratory rate by two device, but is not corrective control respiratory rate. Thus, this study limit to a comparison between two devices.

5 Conclusion

In comparison between capnography using a nasal cannula for continuous monitoring of the respiratory rate, the acoustic method is useful during intravenous general anesthesia in unintubated spontaneously breathing patients undergoing dental procedures. However, the acoustic method might not accurately detect respiratory rate value in cases in which a dental air turbine is used during the procedure.

Acknowledgments The study was supported by departmental funds only. We would like to thank Dr. Takeshi Yokoyama for his help in reviewing the manuscript. This work was supported by the departmental research fund of Kagoshima University.

Author contributions KO participated in the study design, data collection, data analysis, scientific input, and manuscript writing and reviewing. KS participated in the design of the study and reviewing of the manuscript. SF reviewed the manuscript. All the authors read and approved the final manuscript.

Compliance with ethical standards

Conflict of interest The authors have no conflicts of interest relevant to this article to disclose.

References

1. Khader R, Oreadi D, Finkelman M, Jarmoc M, Chaudhary S, Schumann R, Rosenberg M. A prospective randomized controlled trial of two different sedation sequences for third molar removal in adults. J Oral Maxillofac Surg. 2015;73(2):224–31.
2. Ouchi K, Sugiyama K. Required propofol dose for anesthesia and time to emerge are affected by the use of antiepileptics: prospective cohort study. BMC Anesthesiol. 2015;15:34.
3. Sago T, Harano N, Chogyoji Y, Nunomaki M, Shiiba S, Watanabe S. A nasal high-flow system prevents hypoxia in dental patients under intravenous sedation. J Oral Maxillofac Surg. 2015;73(6):1058–64.
4. Mimoz O, Benard T, Gaucher A, Frasca D, Debaene B. Accuracy of respiratory rate monitoring using a non-invasive acoustic method after general anaesthesia. Br J Anaesth. 2012;108(5):872–5.
5. Ramsay MA, Usman M, Lagow E, Mendoza M, Untalan E, De Vol E. The accuracy, precision and reliability of measuring ventilatory rate and detecting ventilatory pause by rainbow acoustic method monitoring and capnometry. Anesth Analg. 2013;117(1):69–75.
6. Autet LM, Frasca D, Pinsard M, Cancel A, Rousseau L, Debaene B, Mimoz O. Evaluation of acoustic method respiration rate monitoring after extubation in intensive care unit patients. Br J Anaesth. 2014;113(1):195–7.
7. Guechi Y, Pichot A, Frasca D, Rayeh-Pelardy F, Lardeur JY, Mimoz O. Assessment of noninvasive acoustic method respiration rate monitoring in patients admitted to an Emergency Department for drug or alcoholic poisoning. J Clin Monit Comput. 2015;29(6):721–6.
8. Marsh B, White M, Morton N, Kenny GN. Pharmacokinetic model driven infusion of propofol in children. Br J Anaesth. 1991;67(1):41–8.
9. Keidan I, Gravenstein D, Berkenstadt H, Ziv A, Shavit I, Sidi A. Supplemental oxygen compromises the use of pulse oximetry for detection of apnea and hypoventilation during sedation in simulated pediatric patients. Pediatrics. 2008;122(2):293–8.
10. Fu ES, Downs JB, Schweiger JW, Miguel RV, Smith RA. Supplemental oxygen impairs detection of hypoventilation by pulse oximetry. Chest. 2004;126(5):1552–8.
11. Cretikos MA, Bellomo R, Hillman K, Chen J, Finfer S, Flabouris A. Respiratory rate: the neglected vital sign. Med J Aust. 2008;188(11):657–9.
12. Soto RG, Fu ES, Vila H, Jr., Miguel RV. Capnography accurately detects apnea during monitored anesthesia care. *Anesth Analg*. 2004;99(2):379–382, table of contents.
13. Coates BM, Chaize R, Goodman DM, Rozenfeld RA. Performance of capnometry in non-intubated infants in the pediatric intensive care unit. BMC Pediatr. 2014;14:163.
14. Gaucher A, Frasca D, Mimoz O, Debaene B. Accuracy of respiratory rate monitoring by capnometry using the Capnomask(R) in extubated patients receiving supplemental oxygen after surgery. Br J Anaesth. 2012;108(2):316–20.
15. Langhan ML. Acute alcohol intoxication in adolescents: frequency of respiratory depression. J Emerg Med. 2013;44(6):1063–9.
16. Patino M, Redford DT, Quigley TW, Mahmoud M, Kurth CD, Szmuk P. Accuracy of acoustic method respiration rate monitoring in pediatric patients. Paediatr Anaesth. 2013;23(12):1166–73.
17. Frasca D, Geraud L, Charriere JM, Debaene B, Mimoz O. Comparison of acoustic method and impedance methods with mask capnometry to assess respiration rate in obese patients recovering from general anaesthesia. Anaesthesia. 2015;70(1):26–31.
18. Atkins JH, Mandel JE. Performance of Masimo rainbow acoustic method monitoring for tracking changing respiratory rates under laryngeal mask airway general anesthesia for surgical procedures in the operating room: a prospective observational study. Anesth Analg. 2014;119(6):1307–14.